RGCDNet: Radiometric–Geometric Collaborative Learning for Change Detection in Remote Sensing

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***Abstract*—In remote sensing change detection (CD), dealing with the interference from pseudochanges such as those caused by illumination variations and geometric registration errors is a key challenge. Integrating physical characteristics of radiation and geometry into a deep network can help to reduce such pseudochanges. As a consequence, this article proposes a novel radiometric–geometric collaborative learning network for CD in remote sensing images (RGCDNet), which incorporates a multisource radiometric shadow suppression (MSR-SS) module and a collaborative registration module (CRM) in order to effectively mitigate the pseudochanges caused by shadows and geometric registration errors. MSR-SS uses the radiometric characteristics of vegetation and water bodies to enhance the separability between shadows and other land features, and constructs spectral separation and suppression processes with a trainable network to minimize false positives (FPs). CRM, meanwhile, achieves collaborative optimization of geometric registration and CD by applying bidirectional deformation to feature maps of various stages and integrating a unified flow field through weighted fusion. Experimental results show that, compared with existing state-of-the-art methods, our RGCDNet significantly improves detection accuracy and robustness. Specifically, we obtained $F1$-scores of 91.31%, 83.32%, and 94.21% on the LEVIR-CD, SYSU-CD, and WHU-CD datasets, respectively. The code of this work can be found on https://github.com/4043-lab/Change-detection**

***Index Terms*—Change detection (CD), collaborative registration module (CRM), multisource radiometric shadow suppression (MSR-SS) module, remote sensing image.**

I. Introduction

THE task of change detection (CD) involves identifying changes in the state of specific objects or phenomena by comparing images captured at different times. Remote sensing image CD, a particular application of this technology, has a wide range of applications in multiple fields, including urban development analysis [1], [2], [3], [4], environmental monitoring, land use management [5], [6], [7], and disaster assessment [8].

In the realm of remote sensing imagery, the CD task has perpetually served as a prominent research focus and has achieved numerous significant accomplishments. However, due to variations in imaging conditions, acquiring multitemporal images may lead to irrelevant interferences, manifested as shadows caused by terrain objects blocking sunlight and misalignments due to changes in imaging perspectives. Consequently, effectively utilizing these images to accurately detect areas of change becomes challenging.

To date, traditional CD techniques, including algebra-based, transformation-based, and classification-based methods, have been extensively researched. However, these traditional techniques are evidently encumbered with conspicuous limitations. Without supervision information, they are preponderantly based on empirical design and are lacking in adequate robustness. This issue becomes more prominent especially when confronted with high-resolution remote sensing images with complex texture features and minute image details. In such cases, they lack the capability to deal with the semantics of changes and understand the scenes. With the semantic learning capability of deep learning, it has become the mainstream method for supervised CD in remote sensing images.

Convolutional neural networks (CNNs), with their powerful feature extraction capabilities, were among the first to be applied to CD, such as FCN and U-Net. Peng et al. [11] proposed the multibranch fusion of U-Net++. Fang et al. [12] combined a Siamese network with NestedUNet to form the SNUNet. Shelhamer et al. [13] proposed three FCN-based methods: "early fusion" FC-Siam-Conc, "late fusion" FC-EF, and FC-Siam-Diff. The "early fusion" feeds the union or difference of bitemporal image pairs into a single-branch network, aiming to directly capture change features but prone to losing details. The "late fusion" inputs single temporal images into a dual-branch (Siamese) architecture for independent learning before fusion, although initially lacking sufficient perception of bitemporal synergy. Subsequently, most techniques use the aforementioned methods to extract features from bitemporal images [1], [14], [15], [16]. However, when dealing with complex scenes, the networks could not distinguish which parts were more important, resulting in resource waste as the model processes large amounts of irrelevant information. To further enhance performance in complex scenes, attention mechanisms have been incorporated. Spatial attention highlights key areas [17], [18], [19]; self-attention explores pixel

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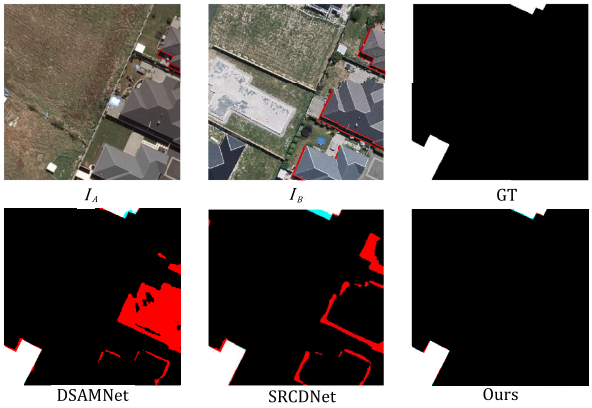


Fig. 1. Impact of shadows on CD results. The first row shows the bitemporal remote sensing images I_A and I_B with shadow areas highlighted by red boxes, along with the GT, where white indicates changed regions and black indicates unchanged regions. The second row displays the results from our method and other methods [9], [10], where red indicates FPs and blue indicates FNs.

changes [20], [21], [22], [23], cross-modal attention [24], and dual-attention fusion approach [25]; and cross-attention fortifies bitemporal pixel correlations [26], [27]. Additionally, leading-edge CD models have employed bilinear interpolation or convolution for multiscale feature registration, aggregating high-level and low-level information to capture both macroscopic and microscopic changes.

Despite the promising results achieved by the aforementioned methods, most of them focus on feature fusion [16] and detail extraction [17], [20], while ignoring the issue of pseudochanges that inherently exist in remote sensing images. In CD, many variances between multitemporal images may cause pseudochanges, among which shadows and geometric registration errors are frequently challenging to deal with. As a consequence, this article aims to reduce the impact of the two types of variances. First, although shadows are a common interfering factor in remote sensing images, their impact on detection results is often overlooked in CD tasks. To effectively mitigate the interference of shadows, it is essential to first accurately detect their presence. Shadow detection methods in remote sensing can be roughly divided into those based on physical characteristics [28], [29], [30] and those based on deep learning [31], [32], [33], [34]. Since the generation of shadows follows basic radiometric mechanisms, many physical characteristics can be modeled, including color space [28], inherent radiometric properties [29], and texture features [30]. Due to the lack of corresponding annotation information and the interpretability of radiometric, MSR-SS is derived from the inherent radiometric properties of different land cover types, such as vegetation and water bodies, to enhance shadow detection and suppression in remote sensing images, as shown in Fig. 1. Second, even though multitemporal remote sensing images have been registered with high precision, geometric registration errors still exist due to viewing angle differences and complex topography of the imaged areas. In this article, we introduce a collaborative registration module (CRM) that integrates image geometric registration into the decoder for synchronized optimization. The radiometry-driven MSR-SS module constructs a spectral separation and suppression process using multisource radiometric features, while the CRM addresses distortions caused by style variations in geometric registration using a unified flow field fusion method. We propose this unified flow field fusion method that mitigates distortions from style changes and enhances CD accuracy. By incorporating both radiometric and geometric properties into the backbone network collaboratively via these modules, they work together to significantly enhance the performance of CD.

The main contributions of this work can be summarized as follows.

1) We propose a novel CD network named radiometric-geometric collaborative learning network for change detection in remote sensing images (RGCDNet). It focuses on the shadows and registration errors which are usual in high-resolution remote sensing images.
2) We design a radiation-driven module that integrates multisource radiometric characteristics for suppressing the shadow impact. We construct the spectral separation and suppression processes via trainable network architecture in order to reduce the likelihood of false positives (FPs) and enhance the overall performance of the CD model.
3) We innovatively incorporate bitemporal image geometric registration into the decoder to achieve coordinated optimization of geometric registration and CD. To address the distortion issue caused by style changes in registration tasks, we propose a unified flow field fusion method that preserves the original image details and reduce false detections in CD.
4) Through a series of experimental comparisons, our method achieves superior performance compared to other state-of-the-art methods on the most canonical and widely used datasets for remote sensing image CD, including the WHU-CD, LEVER-CD, and SYSU-CD datasets.

The remainder of this article is organized as follows. Section II (Related Work) reviews existing methods in CD and shadow detection, discussing their limitations. Section III (Method) details the architecture of our proposed model and its novel algorithm. Section IV (Experiments) describes the datasets and experimental setup, presents a comprehensive comparison with state-of-the-art methods, and provides ablation studies. Finally, Section V (Conclusion) summarizes the main findings, discusses the practical implications of this work, and suggests potential directions for future research.

II. Related Works

A. Change Detection

From a mathematical perspective, the task of remote sensing image CD can be understood as follows: Within the same specific geographical area, we have two remote sensing images acquired at different time points. The goal is to determine whether the ground objects in these two images have changed and to output a binary change map indicating which areas have changed and which have not.

CD methods can be categorized into traditional methods and deep learning-based methods. Currently, deep learning-based methods can be further categorized into image-level

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methods and feature-level methods. The image-level method directly concatenates the bitemporal images along the channel dimension and then inputs them into a single segmentation network for detection. The feature-level method utilizes a pair of weight-shared networks to independently extract features from each single-temporal image.

For example, Shelhamer et al. [13] were the first to propose a fully convolutional network for CD. Subsequently, based on this work, Guo et al. [15] combined the fully convolutional Siamese network with contrast loss to significantly improve performance; Liu and Shi [9] took a different approach and imposed strict constraints on the differential features of the output at each stage of the encoder to generate a finer representation; Li et al. [22] proposed a dense skip connection module to efficiently realize multilevel feature aggregation between the encoder and the decoder.

Some cutting-edge methods keenly captured the shortcoming of CNN in capturing long-distance dependencies and innovatively introduced the Transformer module. Zhang et al. [35] used many Swin Transformer blocks to construct a pure Transformer Siamese CD network; Feng et al. [20] ingeniously proposed a novel method for feature fusion using the Transformer module, using two weight-shared backbones to extract unique features from bitemporal respectively, and then promotes information aggregation between different backbone features and scale features based on single-temporal features; Chen et al. [36] proposed a BIT composed of one Transformer encoder and two Transformer decoders to comprehensively capture temporal and spatial context information.

Recently, Luo et al. [1] have proposed a multiscale diff-changed feature fusion network. This network enhances the feature representation capability by learning the fine-grained change components of bitemporal hyperspectral images at various scales, enabling better detection of subtle changes.

The aforementioned methods have made certain achievements in either the feature extraction or fusion aspects, successfully improving the ability to detect the real changes of ground objects in remote sensing images. However, most of them have overlooked the influence of pseudochanges on the results. In this article, we mainly focus on pseudochanges caused by shadows and registration errors which commonly exist in high-resolution remote sensing images.

B. Shadow Detection

In remote sensing images, shadows refer to the dark areas that result from the obstruction of sunlight or sensor signals by objects such as terrain undulations, buildings, and vegetation. These areas cannot directly receive sunlight or sensor signals, making them appear darker in images. Shadows are one of the highly interfering factors that may cause pseudochanges in remote sensing image CD, and the prerequisite for suppressing shadows is detecting shadows.

Because the shadows are usually with lower brightness values, the intuitive way is to segment the shadows using threshold-based methods [37], [38], but not all dark regions are shadows. Therefore, geometric optical models [28], [39] were proposed. These types of methods attempt to accurately simulate the geometric relationship among the light source, ground objects, and shadow. They calculate the theoretical position and shape of the shadow through known parameters such as the solar altitude angle, the azimuth angle, and the height of the ground objects.

The vigorous development of deep learning has brought new ideas for shadow suppression. By training an end-to-end deep network [31], [32], [40] via a large amount of annotated data, the network can learn complex features such as the texture and brightness gradient of the shadows, and then output the shadow segmentation map. Generally speaking, both traditional physical models and deep learning methods have made progress in the field of shadow suppression, but for CD, there are few methods focusing on reducing the pseudochanges caused by shadows.

Without corresponding annotation information of shadows in CD problems, we try to use differences in spectral radiance characteristics to separate shadows. Liu et al. [41] observed that vegetation typically exhibits low reflectance in visible light bands but appears bright in the near-infrared (NIR) band, while inland water bodies demonstrate higher reflectance in the green band than in the blue band. These distinct physical properties enhance the separability between shadows and interfering objects, thereby enabling effective shadow extraction. Inspired by this approach, we innovatively adapt this physical process into a deep learning framework by proposing a multisource radiometric shadow suppression (MSR-SS) module, which enhances shadow detection and suppression through multisource radiometric feature integration. Specifically, we replace traditional spectral bands with deep feature maps, leveraging neural networks' representation learning capability to simulate the physical shadow separation process. More importantly, with CD supervision signals, our model learns to automatically differentiate between task-relevant and irrelevant shadow regions, effectively reducing interference from unrelated shadows. Compared to existing methods, our MSR-SS incorporates the physical shadow separation process into a deep learning framework for the first time, specifically designs a shadow suppression mechanism for CD tasks, and addresses the lack of dedicated shadow suppression methods in this field.

C. Registration

Research on feature-based image registration involves geometric features including surface features, linear features, and point features. In practical applications, point features are more commonly used than surface and linear features because they are easier to extract, locate, and describe [42]. Traditional feature extraction methods, such as the SUSAN [43] corner extraction operator, SIFT [44], and RIFT [45], mostly rely on the shallow information of the image to perform the tasks of feature point extraction and description [46].

Deep learning-based methods are usually designed following the two mainstream strategies, i.e., feature-based approaches and end-to-end learning approaches. The first strategy is based on feature extraction. Such methods generally output score maps and then accurately extract the points with the largest local response values as feature points. TILDE [47] and SuperPoint [48] exemplify this approach. TILDE selects

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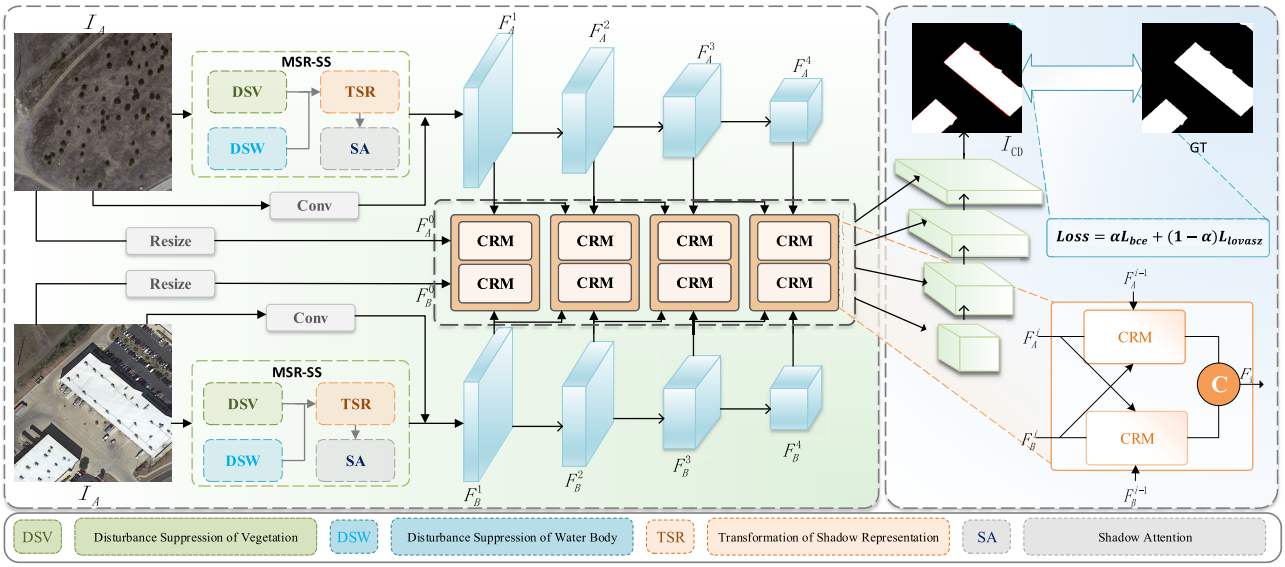


Fig. 2. Overall architecture of the proposed RGCDNet. The network takes bitemporal remote sensing images as input. It first enhances shadow-related cues via the MSR-SS to generate shadow-attention maps. These attention maps are fused with the initial convolutional features extracted from the original images to obtain shadow-corrected enhanced features. The enhanced features are then fed into an encoder to extract multiscale representations $\{F_A^i\}_{i=1}^4$ and $\{F_B^i\}_{i=1}^4$. Next, the CRM performs geometric registration by predicting deformation fields and applying bidirectional warping with weighted fusion, producing the registered feature pyramid $\{F_R^i\}_{i=1}^4$. Finally, the CD decoder integrates the shadow-corrected features with the registered features to generate the final change map I_{CD}.

image patches containing stable feature points as training samples, conducts strict supervised training on the regressor, and efficiently extracts image features based on the output feature response intensity map. To effectively address the deficiencies of TILDE [47], subsequent SuperPoint [48] and Quad-networks [49] innovatively proposed using synthetic datasets and spatial correspondence-based loss functions to enhance the network's ability to handle geometric distortions. The second strategy is based on end-to-end learning. Typical algorithms in this category include KeyNet [50], DeepReg [51], and cycleGAN [52]. The KeyNet is an end-to-end learning approach that leverages CNNs to generate score maps and extract features across different scales, achieving robust feature extraction by integrating traditional Harris corner features with deep learning.

Recent studies have been dedicated to improving the architecture and training algorithm of the deep learning model. Some studies have proposed a method of joint image registration and segmentation (JRS) [53], [54], [55], [56]. Such methods tend to construct two independent encoder–decoder structures to perform registration and segmentation tasks, respectively. In contrast, our CRM innovatively integrates geometric registration and CD into a single unified framework, leveraging bidirectional deformation fields to precisely align images and ensure feature matching, while employing a weighted fusion strategy to dynamically adjust image weights, thereby enhancing the robustness and accuracy of CD.

III. Method

In this section, we introduce the proposed RGCDNet from the perspectives of architecture, modules, and objectives. The overall architecture is shown in Fig. 2. Our network is designed to address two major pseudochanges: shadows and geometric registration errors. To mitigate the impact of shadows on the CD task, the MSR-SS module is integrated into the encoder. This module is to delineate the shadow regions that could interfere with CD, thereby preventing false detections in these regions. Regarding the geometric registration errors, the CRM constructs a dual-branch decoder to address this issue, which jointly optimizes the geometric alignment and CD tasks. Finally, the features from the shadow suppression encoder and the geometric registration decoder are fed into the CD decoder to integrate the composite features and thus generate the change map. The following is our detailed explanation of the proposed model.

A. Overview

The proposed RGCDNet architecture is illustrated in Fig. 2. Prior to elaborating on the specific details of the network, we first expound the overall workflow of employing our proposed method to carry out the CD task. Let the input multitemporal remote sensing image pairs be denoted as $I_A^{3\times H\times W}$ and $I_B^{3\times H\times W}$, where W and H respectively denote the width and height of the image pairs.

First, the bitemporal features are fed into two MSR-SS modules S_d respectively, then the shadow feature attention maps S_A and S_B are acquired

$$S_A = S_d(I_A), S_B = S_d(I_B). \tag{1}$$

These attention maps can reflect the feature distribution and corresponding attention level of the shadow area in the image. Subsequently, the shadow attention maps are applied to the feature maps, and the resultant shadow correction enhanced features then pass through four downsampling layers to extract features of different scales

$$\{F_A^i\}_{i=1}^4 = Encoder\left(F_{\text{init}}^A + \lambda S_A\right)$$

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$$\left\{F_B^i\right\}_{i=1}^4 = Encoder\left(F_{\text{init}}^B + \lambda S_B\right) \tag{2}$$

where λ is a learnable parameter and F_{init} denotes the features extracted by the initial convolution layer; $(I_A + \lambda S_A)$ and $(I_B + \lambda S_B)$ are the shadow correction enhanced images; and $\{F_A^i\}_{i=1}^4$ and $\{F_B^i\}_{i=1}^4$ are the features of four different dimensions. The feature maps are extracted using a Siamese network structure, where both branches share the same weights. The network consists of a series of convolutional layers, similar to the encoder part of UNet++ [57], that process the two input images I_A and I_B independently to capture spatial and temporal features. The shared weights ensure that both branches learn similar feature representations from the two input images, which are then used for generating shadow attention maps and for the subsequent CD process. Subsequently, the features of different scales mentioned above and the original bitemporal image pairs are fed into the geometric registration decoder. The decoder can integrate the feature information from diverse scales to generate a flow field. This flow field essentially reflects the mapping relationship of the image pairs in spatial positions and other aspects, playing a crucial role in subsequent precise geometric registration and correction. Then, by combining the generated flow field with the original image pairs, the decoder exploits the mapping information to correct them and generate the registered and corrected features as follows:

$$\left\{F_R^i\right\}_{i=1}^4 = Reg_Decoder\left(I_A, I_B, \left\{F_A^i\right\}_{i=1}^4, \left\{F_B^i\right\}_{i=1}^4\right). \tag{3}$$

Finally, the features that have undergone correction of shadow and geometric registration errors are fused with the features of different scales, so as to achieve the purpose of suppressing two specific pseudochanges. Subsequently, the change decoder will convert these features into binary images, and these binary images will be output as the final CD results as follows:

$$I_{CD} = Change_Decoder\left(\left\{F_A^i\right\}_{i=1}^4, \left\{F_B^i\right\}_{i=1}^4, \left\{F_R^i\right\}_{i=1}^4\right). \tag{4}$$

Regarding the CD decoder, we designed a network architecture based on a multiscale feature pyramid. The decoder takes bitemporal features from multiple scales as input, together with the registered feature pyramid $\{F_R^i\}_{i=1}^4$ produced by the geometric registration decoder. At each scale, the decoder fuses spatiotemporal information by concatenating the corresponding features along the channel dimension. Specifically, $\{F_R^i\}_{i=1}^4$ is concatenated with the bitemporal features at the same scale to provide spatial alignment cues that help alleviate residual misalignment between the two temporal inputs. The decoder then employs a top-down pathway with progressive upsampling to restore spatial resolution. Finally, a 1×1 convolution layer followed by a Sigmoid activation function produces the change probability map with the same spatial dimensions as the input images. In this way, the CD decoder leverages both the bitemporal feature representations and the alignment information carried by $\{F_R^i\}_{i=1}^4$ to accurately detect changes between the two input images.

B. Multisource Radiational Shadow Suppression Module

During the acquisition process of remote sensing images, shadows are often formed due to the obstruction of ground objects by light. The appearance of shadows will introduce pseudochanges into the images, which subsequently leads to information loss and distortion, significantly increasing the complexity of CD and highly prone to false detection results. Through extensive literature review, we have found that, in remote sensing images, features such as vegetation and inland water bodies are often confused with shadows. This is mainly attributed to the fact that such features mostly exhibit low luminance characteristics, and their reflectance levels are quite similar to those of shadows. Therefore, effectively distinguishing among vegetation, inland water bodies, and shadows has become the core task of the MSR-SS.

Multisource radiation refers to the radiation characteristics exhibited by various different ground objects. In our module, it mainly refers to the inherent radiation characteristics of the ground objects that interfere with shadow detection, namely, those of vegetation and inland water bodies. For vegetation, its reflectance is similar to that of shadows in the visible light band. However, in the NIR band, due to the special reflection mechanism of the internal cell structure of vegetation leaves for NIR light, it has a high reflectance, which is significantly different from that of shadows. As for inland water bodies, their overall reflectance is relatively low in both the visible and infrared bands, but the reflectance of inland water bodies in the green band is higher than that in the blue band. In contrast, a shadow is formed by an object blocking direct sunlight, and its main light source is skylight. Skylight comes from the scattering of sunlight, according to the Rayleigh scattering law: The relationship between the intensity of light and its wavelength is described by the following formula:

$$I(\lambda) \propto \frac{1}{\lambda^4} \tag{5}$$

where $I(\lambda)$ represents the intensity of light with a wavelength of λ. It can be known that light with short wavelengths, such as blue light, has a high scattering intensity. Therefore, the shadow area has a relatively high reflectance in the blue band, which is higher than that in the green band. The characteristics of various ground objects in different bands can enhance the separability of shadows. The CD dataset only contains images with three RGB channels. We first perform convolution on the R, G, and B channels separately to generate the corresponding feature maps R', G', and B', which contain the information of the three bands respectively. We estimate the NIR channel from RGB images by exploiting the spectral correlation between visible and NIR bands in vegetation. Using the Visible-Band Difference Vegetation Index (VDVI) [58] computed from RGB channels, we derive an initial NIR estimate that preserves the physical constraint of NIR >R reflectance. To enhance the estimation accuracy, we construct a lightweight CNN to optimize this derivation process. First, we accurately compute the VDVI using 1×1 convolutions with fixed weights. Then, we convert VDVI into an initial NIR estimate that complies with the physical constraint, ensuring that NIR reflectance is always higher than the red channel. Finally, we refine this initial estimate through a lightweight convolutional network to generate the final high-quality NIR feature representation $\mathbf{C}$.

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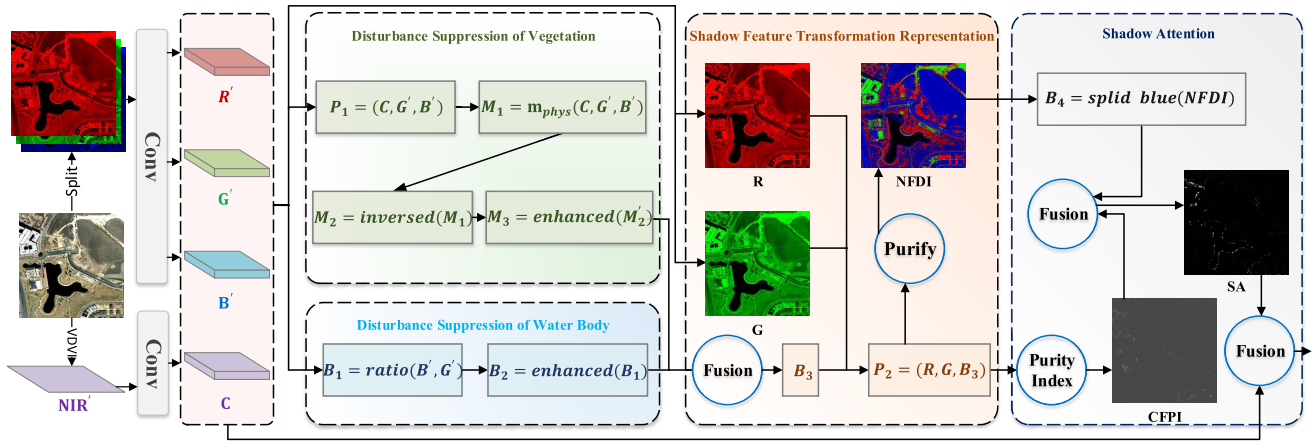


Fig. 3. Architecture of the proposed MSR-SS module. The MSR-SS module enhances shadow features in bitemporal remote sensing images by suppressing interference from vegetation and water bodies. It includes four key steps: 1) vegetation interference suppression using NIR reflectance; 2) water body interference suppression using blue-green band reflectance ratios; 3) shadow feature transformation to enhance shadow separability; and 4) shadow attention to generate a shadow-focused feature map.

We design a deep learning method for shadow separation based on the aforementioned multisource radiation characteristics, namely MSR-SS. The overall flowchart is shown in Fig. 3. Next, we will describe this module in detail in four parts: vegetation interference suppression, water body interference suppression, shadow transformation representation, and shadow separation.

1) Vegetation Interference Suppression: Vegetation has a high reflectance in the NIR band, while shadows have a relatively low reflectance. Based on this characteristic, shadows and vegetation can be separated. Therefore, we simulate the separation process via a trainable network layer, specifically a convolutional layer, to generate features. Then, based on the physical characteristics of different ground objects, we perform shadow inference. To fully leverage the high reflectance of vegetation in the NIR band, a false-color composite feature map P_1 is generated by integrating the C, G', and B'. The features G' and B' are derived through convolutional operations to enhance the spectral characteristics in the green and blue bands most relevant for shadow-vegetation separation, while avoiding the numerical instability associated with raw pixel operations. The component C is estimated to simulate an NIR band, which is crucial for highlighting vegetation (high NIR reflectance) and suppressing its confusion with shadows. Subsequently, we converted the P_1 image into a grayscale image to extract its intensity information

$$M_1 = max_{phys}\left(C, G', B'\right) \tag{6}$$

where $max_{phys}(\cdot)$ is a differentiable approximation of the max operation implemented through a three-layer CNN. The network processes stacked input feature maps through convolutional blocks with Sigmoid activations, learning to generate spatial weight distributions that emphasize dominant features while preserving spatial relationships. This weighted summation approach effectively simulates physical-band selection behavior while maintaining gradient flow for end-to-end training. The same architecture can be adapted for min operations by inverting the selection logic.

This method contributes to enhancing the separability between shadows and vegetation, which is manifested by the fact that vegetation shows high brightness in M_1, while shadows maintain low brightness. Subsequently, an inversion operation is performed on the feature map M_1

$$M_2 = inversed\,(M_1) = 2^N + Neg\,(M_1) - 1 \tag{7}$$

where N is the quantization bits of the image. For instance, in the case of an 8-bit image, $N = 8$, and the function $Neg(\cdot)$ represents negation. According to this formula, M_2 is obtained, and at this time, the shadow regions seem brighter. However, different images possess certain discrepancies. Therefore, we further adopt the method of percentage linear stretching (a technique used to enhance the contrast of an image by stretching its intensity range based on a specified percentage of the data distribution, typically excluding the lowest and highest few percentiles to avoid oversaturation and undersaturation) to perform normalization processing on M_2 at a rate of 2%, thereby obtaining M_2'.

In order to enhance the difference between shaded and nonshaded areas, gamma correction is introduced as follows:

$$M_3 = enhanced\left(M_2'\right) = \left(M_2'\right)^{\gamma} \tag{8}$$

$$\gamma = \begin{cases} \dfrac{2v_m}{T} - 1, & \text{if } v_m \geq T \\ \dfrac{2v_m}{T}, & \text{otherwise} \end{cases} \tag{9}$$

where T is a learnable parameter, and v_m represents the mean value of the image M_2'. By calculating the gamma value, the brightness of nonshaded areas can be reduced while maintaining high brightness in shaded areas, thereby effectively suppressing vegetation interference and enhancing shadow features.

2) Water Body Interference Suppression: Based on the characteristics that the reflectance of shadows is relatively high in the blue band and the reflectance of inland water bodies is higher in the green band than in the blue band, the ratio of

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the green-band feature map G' to the blue-band feature map B' is defined as follows:

$$B_1 = ratio\left(B', G'\right) = \frac{2}{\pi}\arctan\left(B'/G'\right). \tag{10}$$

The expression $2/\pi$ is used to normalize the result of arctan$(\cdot)$ to the range of $[0, 1]$. In the ratio image, water bodies appear dark, while shadows appear bright. For convenience in subsequent processing, B_1 is first normalized using the same normalization method as M_2'. Then, using the aforementioned gamma correction method, the image B_2 is obtained, which significantly reduces the brightness of the water bodies and helps to distinguish between the water bodies and the shadows.

3) Shadow Feature Transformation Representation: After the above processing, the features of the shaded areas in the M_3 and B_2 images are significantly enhanced. In the method of extracting shadows using the binary feature map, its accuracy heavily depends on the precision of the selected threshold. Therefore, we propose to adopt the color model to represent shadows and perform the fusion operation on M_3 and B_2

$$B_3(x, y) = Fusion(M_3, B_2) = (M_3 \otimes B_2) / \left(2^N - 1\right) \tag{11}$$

where $\otimes$ is elementwise feature multiplication, and $N = 8$ is the bit depth. This operation replicates the brightness-color cast coupling effect of shadows in physical imaging within the feature space: High-response regions: Areas jointly enhanced by M_3 and B_2 (simulating dominant physical features of shadows); and Low-response regions: Nonshadow features are suppressed via multiplication (simulating the degradation of physical properties in illuminated areas). Subsequently, the false-color feature space P_2 is synthesized by combining B_3 with the original R and G channels. At this time, in the physical process, the candidate shadow appears blue. Meanwhile, in the feature space, the candidate shadow features are prominent. To further enhance the distinction between shadow features and other features, we introduce the concept of feature purification, represented by the Normalized Feature Difference Integrated Index (NFDI). This index quantifies the relative differences among feature channels to highlight the unique characteristics of shadow regions, thereby improving the separability of shadows from nonshadow areas. The NFDI formula is defined as follows:

$$\mathrm{NFDI}^m = \begin{cases} \dfrac{\max_{phys}\left(p^k\right) - \min_{phys}\left(p^k\right)}{\max_{phys}\left(p^k\right)}, & \text{if } m = \underset{k}{\operatorname{argmax}}\ p^k \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $m, k \in \{red, green, blue\}$, $p \in P_2$, p^k is the kth feature channel in the feature map P_2, and $max_{phy}(\cdot)$ is the physical process simulators in the feature space, as previously mentioned. $min_{phys}(\cdot)$ is a learnable approximation of the minimum operation that simulates physical radiation absorption in feature maps. The structure of $min_{phys}(\cdot)$ is essentially the same as that of $max_{phy}(\cdot)$. The NFDI contains only the three primary color feature channels, and the blue feature channel highly corresponds to the original shadow position, validating the equivalence between feature operations and physical process simulation.

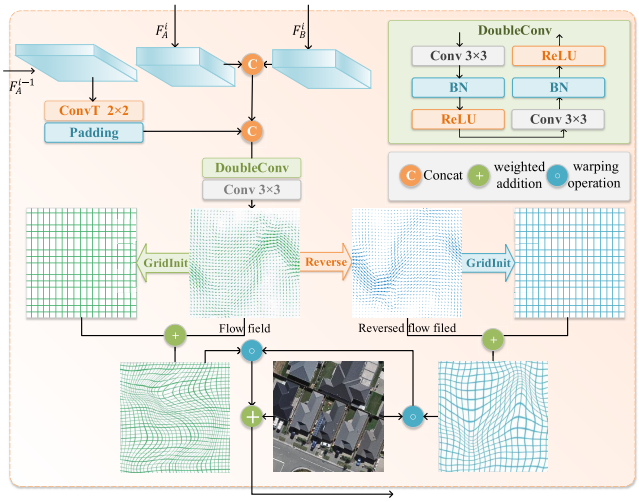


Fig. 4. Architecture of the proposed CRM. CRM predicts deformation fields from dual-temporal feature maps and the previous layer's feature map, performs bidirectional deformation, and fuses flow fields to generate aligned feature maps for CD.

4) Shadow Attention: The candidate shadow feature map B_4 is extracted from the NFDI by isolating the dominant suspected blue-band features, which physically correspond to the characteristic spectral response of shadow regions. To better simulate how shadows selectively absorb and reflect light across different wavelengths, we develop the Color Feature Purity Index (CFPI) as a physical process model

$$\mathrm{CFPI} = 1 - \frac{mid_{phys}\left(P_2^k\right) - min_{phys}\left(P_2^k\right)}{max_{phys}\left(P_2^k\right) - min_{phys}\left(P_2^k\right)} \tag{13}$$

where $\mathrm{mid}_{\mathrm{phys}}(\cdot)$, $\min_{\mathrm{phys}}(\cdot)$, and $\max_{\mathrm{phys}}(\cdot)$ are the trainable physical simulators for spectral median, minimum absorption, and maximum reflectance respectively, as implemented in (6). Finally, the shadow attention map SA is obtained as follows:

$$\mathrm{SA} = \sqrt{B_4 \otimes \mathrm{CFPI}} \tag{14}$$

where pixels in nonshadow regions are set to zero, while those in shadow regions hold positive values. Additionally, a higher pixel value corresponds to a greater confidence level. We will utilize the obtained mask, after final processing, as the attention weight for multisource perception of shadow features. Ultimately, by augmenting the weight of the shadow in this manner, the network designed for remote sensing image CD will be able to detect the presence of the shadow, thereby learning about the influence of the shadow, and finally attaining the effect of intelligent shadow suppression.

C. Collaborative Registration Module

In dual-temporal images, the issue of ground object feature registration errors may arise. In light of this, the CRM has been designed. The proposed CRM is illustrated in Fig. 4. CRM is composed of two decoders, namely the geometric registration decoder and the CD decoder. The geometric registration decoder is tasked with predicting the spatial transformation between images, i.e., the deformation field. Meanwhile, the

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CD decoder serves the purpose of predicting the changed regions by integrating the geometric registered images with the original features. The geometric registration decoder, as the upper-level task, has its optimization dependent on the solution of the CD decoder (the lower-level task). In each iteration, the CD decoder responds to the gradient of the geometric registration decoder, thereby achieving collaborative optimization of the two tasks.

In the dual-level optimization framework, geometric registration, as an auxiliary learning process, can generate a deformation flow field ψ_i from F_A^i to F_B^i. Subsequently, the feature map F_B^i is warped by the deformation field to produce a pseudo-aligned feature map $\hat{F_B^i}$. However, the appearance disparity between F_A^i and F_B^i may lead to a mismatch between the warped $\hat{F_B^i}$ and the original F_B^i. We propose a consistent flow field fusion method to generate robust pseudo-aligned feature maps and employ bidirectional deformation to avoid the aforementioned situation. Specifically, we first predict the deformation field ψ_i from F_A^i to F_B^i, and incorporate the features from the previous layer to enhance the feature representation when predicting the deformation field for each layer. Then, we reregister to obtain the reverse deformation field, thereby obtaining the pseudo-F_A^i feature map. Considering the noise issue of the pseudo-F_A^i feature map, we adopt weighted fusion to combine the deformed image with the undeformed features.

The deformation field ψ_i is computed using the geometric registration decoder as follows:

$$\psi_i = f_{rd}\left(F_A^i, F_A^{i-1}; F_B^i\right) \tag{15}$$

$$\psi_i^{-1} = f_{rd}\left(F_B^i; F_A^i, F_A^{i-1}\right) \tag{16}$$

where $i \in \{1, 2, 3, 4\}$. When $i = 1$, the feature map F^0 is used, which is obtained by resizing the original images I_A and I_B to match the dimensions of F_A^i and F_B^i

$$\hat{F_B^i}' = \theta \cdot \left(F_B^i \circ \psi_i^{-1}\right) \circ \psi_i + (1-\theta) \cdot F_B^i \tag{17}$$

where θ is a random scalar used to adjust the proportion between the reference feature and the aligned feature, $\circ$ denotes the spatial transformation operation that resamples a feature map into the target space according to a deformation field, and ψ_i represents the flow field predicted by the geometric registration decoder. This formulation first uses the inverse deformation field to warp the feature map F_B^i into the feature-space coordinate system of F_A^i; it then applies the forward deformation field to map the inversely warped feature back to the feature-space coordinate system of F_B^i. With the introduction of θ, the original feature and the bidirectionally deformed counterpart are mixed, and the two parts are finally combined to obtain the aligned pseudo-F_B^i feature. This process not only achieves spatial alignment of multiscale features, but also preserves the semantic information and distribution characteristics of the original F_B^i features.

D. Loss Function

In CD tasks, traditional loss functions, such as binary cross-entropy (BCE), mainly concentrate on the accuracy of pixel-level classification and are unable to effectively address the issue of class imbalance between changed pixels and unchanged pixels. Therefore, we combine the cross-entropy loss with the Lovasz-Softmax loss function [59] to remedy this deficiency. BCE is used as the primary loss function

$$L_{bce} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(p_i\right) + \left(1-y_i\right)\log\left(1-p_i\right)\right] \tag{18}$$

where N represents the number of samples, y_i is the true label of the ith sample ($y_i \in \{0, 1\}$), and p_i is the probability that the ith sample is predicted as the positive class.

Let $u \in \mathbb{R}^{H\times W\times C}$ be the model's output, representing the probability distribution of each pixel belonging to different classes before the argmax operation, and $y \in \{0, 1\}^{H\times W\times C}$ be the ground truth (GT) labels in one-hot encoding form.

The marginal probability $m_c(i)$ for pixel i of class c is calculated as follows:

$$m_c\left(i\right) = \frac{1}{|\partial\Omega_i|}\sum_{j\in\partial\Omega_i} u_c\left(j\right) \tag{19}$$

where ∂ is the boundary symbol from set theory, $\partial\Omega_i$ is the neighborhood of pixel i, $|\partial\Omega_i|$ is the cardinality of the neighborhood, and $u_c(j)$ represents the unnormalized prediction (logit) for class c at pixel j. Then, the Lovasz-Softmax loss function L_{lovasz} is defined as follows:

$$L_{lovasz} = \sum_{c=1}^{C}\Delta J_c\left(m_c\right) \tag{20}$$

where $\Delta J_c(m_c)$ is the Lovasz extension based on m_c, which is a function of the marginal probability m_c and measures the difference between the predicted probability and the GT labels.

The combined loss function L using linear combination is given by

$$L = \alpha L_{bce} + (1-\alpha)\, L_{lovasz} \tag{21}$$

where α is a hyperparameter that balances the contribution of L_{bce} and L_{lovasz}.

IV. Experiments

A. Data Description

In this experiment, three remote sensing image CD datasets commonly used in CD tasks were adopted, namely the Building CD Dataset (LEVIR-CD) [2] from the Laboratory for Learning Vision and Remote Sensing, the Wuhan University Dataset (WHU-CD) [60], and the Sun Yat-sen University dataset (SYSU-CD) [61], to conduct comparative and ablation experiments. Due to the illumination and view angle variances, image pairs in the three datasets contain many inferences of shadows and geometric registration errors.

1) LEVIR-CD Dataset: This dataset consists of 673 pairs of high-resolution (0.5 m per pixel) image patches from Google Earth, with each pair having a size of 1024 × 1024 pixels. In our experiments, we use the nonoverlapping cropped patches of 256 × 256, having default data split of train, validation, and test equal to 7120, 1024, and 2048, respectively. The time span between the images ranges from 5 to 14 years, featuring significant land use changes and prominent development in the construction industry. Geospatially, the dual-temporal images

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are sourced from 20 different regions in the state of Texas, United States, such as Austin and Buda. However, CD models may produce FPs or false negatives (FNs) at building edges, as the features of edge pixels are less noticeable due to illumination and shadow effects.

2) SYSU-CD Dataset: The dataset contains 20 000 pairs of aerial images with a size of 256 × 256 and a resolution of 0.5 meters, taken in the Hong Kong region from 2007 to 2014. It covers a variety of change types, including new urban construction, urban expansion, abandoned construction sites, vegetation changes, road expansion, and offshore construction. The SYSU-CD dataset is complex in scenes, focusing not only on changes in buildings but also on changes in other land features such as rivers, water surfaces, and forests. This diversity and complexity also pose significant challenges for CD algorithms.

3) WHU-CD Dataset: This dataset is an aerial and satellite image dataset containing building samples. It includes aerial images from Christchurch, New Zealand, with a resolution of 0.075 m, covering an area of 450 km^2 and containing over 220 000 individual buildings. It then includes satellite images with two subsets: one integrating multisource remote sensing images, and the other covering an area of 550 km^2 in East Asia (with a resolution of 2.7 m). The dataset focuses on the reconstruction areas after the earthquake in 2011. The core of the dataset is the aerial images taken in 2012, covering an area of 20.5 km^2 with 12 796 buildings. The accuracy of these images has been corrected to 1.6 pixels using 30 ground control points (GCPs). The primary challenges of this dataset are its diverse types of land cover, large variations in the scale of changes, and limited annotated data, which collectively increase the difficulty of training and generalizing CD models.

B. Experimental Setup

1) Implementation Details: The experiment was run on a workstation equipped with an NVIDIA RTX 3090 GPU. The PyTorch 2.1.1 deep learning framework was used for model construction and training, and CUDA 11.8 was combined to achieve GPU acceleration. In the setting of hyperparameters, the AdamW optimizer was adopted, and the initial learning rate was set to 0.0001 to balance the convergence speed and stability. The optimizer employed the stochastic gradient descent (SGD) algorithm with a momentum of 0.9. The weight decay coefficient was set to 0.0005, and the number of iterations (iters) was set to 240 000. The model weights were saved every ten iterations so that it would be possible to roll back to a better model in case of overfitting. Additionally, the hyperparameter α in the composite loss function was set to 0.5, as determined through comparative experiments. This value was used consistently across all experiments and was found to provide the best performance across multiple datasets.

2) Evaluation Metrics: In this experiment, we used multiple evaluation metrics to assess the model's performance. While Recall and Precision are standard metrics, they can be somewhat misleading in cases of unbalanced classes. Therefore, we primarily refer to intersection over union (IoU), $F1$-score, and Kappa coefficient for a more robust evaluation. Here, TP represents true positives, FN represents false negatives, FP represents false positives, and TN represents true negatives.

Recall Rate: Recall measures the model's ability to find actual changes. The formula is

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{22}$$

Precision Rate: Precision indicates the proportion of correct changes among predicted changes. The formula is

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{23}$$

Overall Accuracy (OA): OA shows overall prediction accuracy. The formula is

$$OA = \frac{TP + TN}{TP + TN + FP + FN}. \tag{24}$$

$F1$-Score: $F1$-score is the harmonic mean of precision and recall. The formula is

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{25}$$

IoU: IoU measures the overlap of predicted and actual changes. The formula is

$$IoU = \frac{TP}{TP + FP + FN}. \tag{26}$$

Kappa Coefficient: Kappa measures classification consistency. The formula is

$$\text{Kappa} = \frac{p_o - p_e}{1 - p_e} \tag{27}$$

where p_o is the actual classification accuracy (equivalent to OA), and p_e is the expected accuracy of random classification.

C. Comparison With State-of-the-Art Methods

1) FC-Siam-Diff [14]: Similarly, two identical network branches are utilized to handle two input images separately. In the decoding stage, the absolute differences of the skip connections of the two branches are concatenated.

2) FC-Siam-Conc [14]: Two identical network branches are employed to process two input images respectively, with each branch sharing the same weights. During the decoding stage, the skip connections of the two branches are concatenated.

3) SRCDNet [10]: It is a network designed for CD in images with different resolutions. The low-resolution images are first enhanced to high resolution through the super-resolution module. Subsequently, the Siamese network structure and the attention mechanism are adopted to extract features and conduct CD.

4) DSAMNet [9]: The Siamese network structure is exploited to extract multiscale features, and the discriminability of the features is enhanced via metric learning and the convolutional block attention module (CBAM). Moreover, the introduction of the deep supervision technique further improves the learning ability of the model. Eventually, a more accurate CD is achieved by optimizing the training process with the hybrid loss function.

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5) Bit [23]: The dual-temporal images are represented as a small number of visual words, and the Transformer encoder is used to model the context within a compact token space-time. Then, the learned context-enriched tokens are fed back to the pixel space through the Transformer decoder. Finally, the feature difference image is calculated and input into a shallow CNN to generate pixel-level change predictions.

6) MSCANet [62]: It combines the advantages of CNN and Transformer. The CNN-based feature extractor captures hierarchical features. Then, the multiscale context aggregator (MSCA) within the Transformer architecture encodes and aggregates context information. Finally, a multibranch prediction head (MBPH) is adopted to obtain the change map, enhancing the learning of deep layer features.

7) DDCNN [17]: It is composed of convolutional units, upsampling attention units (UA), and difference enhancement units (DE). The convolutional units are utilized for feature extraction. The UA units are employed to enhance texture and detail information during the upsampling stage, while the DE units are used to process the difference maps of dual-temporal images.

8) DMINet [20]: Its network architecture consists of two parallel Siamese pretrained CNNs. The dual-temporal features are interacted through the temporal joint attention module. Two strategies, namely pixel-level subtraction and channel-level concatenation, are adopted to obtain the difference features. Through a simple Hadamard product operation, the difference features at different levels are gradually aligned.

9) AANet [63]: Employing ResNet-18 as the backbone network, to resolve the ambiguous region issue in pseudochange problems, we introduce the ambiguity refinement module (ARM). ARM is composed of two submodules: the ambiguity and difference extractor (ADE) and the ambiguity-guided difference refiner (AGDR). The ADE serves to extract ambiguous regions and difference features, and the AGDR refines these features using the information obtained by ADE.

10) HRMNet [64]: HRMNet is a novel deep learning framework for remote sensing CD that boosts accuracy on hard regions—such as change boundaries, tiny objects, and pseudochanges—via a cross-layer knowledge review module and an online hard-region estimation branch.

11) ScartchFormer [65]: ScratchFormer is an end-to-end Transformer-based method for remote sensing CD that does not rely on pretrained models. By virtue of two major innovations—"Shuffled Sparse Attention (SSA)" and "Change-Enhanced Feature Fusion (CEFF)"—it breaks through the "pretraining dependence" bottleneck of existing Transformer-based CD methods.

D. Comparing Experiments

In this section, we comprehensively evaluated the performance of multiple methods introduced in the previous section, including our proposed approach, on three different datasets, using a combination of quantitative and qualitative analyses. For a focused and clear comparison, the subsequent qualitative figures will highlight the most competitive contemporary methods, while omitting the earlier FC-Siam-Diff and FC-Siam-Conc baselines. Their results are included in the quantitative tables to provide a complete numerical overview of all compared techniques. Quantitative results are presented in tables, with relevant metrics already described in the "Evaluation Metrics" section. Qualitative results are shown directly via images. In these images, white denotes True Positive (TP, correctly identifying positive samples as positive), red represents False Positive (FP, misclassifying negative samples as positive), black indicates True Negative (TN, correctly identifying negative samples as negative), and cyan stands for False Negative (FN, misclassifying positive samples as negative). Each dataset is associated with one table and one image, and detailed descriptions will be given for each dataset hereinafter.

TABLE I

COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON LEVIR-CD DATASET. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND GREEN

Methods	Rec(%)	Pre(%)	OA(%)	F1(%)	IOU(%)	KC(%)
FC-Siam-diff	83.16	90.05	98.67	86.47	76.16	85.77
FC-Siam-conc	84.66	89.14	98.69	86.84	76.75	86.16
SRCDNet	90.71	81.49	98.48	85.85	75.21	85.05
DSAMNet	89.09	80.23	98.33	84.43	73.05	83.54
BIT	86.98	92.16	98.96	89.50	80.99	88.95
MSCANet	89.10	90.41	98.96	89.75	81.40	89.20
DDCNN	89.08	92.18	99.06	90.60	82.82	90.10
DMINet	89.00	92.64	99.08	90.78	83.12	90.30
AANet	90.25	91.29	96.10	90.77	83.13	88.30
HRMNet	88.69	90.53	98.99	89.92	81.69	89.39
ScratchFormer	90.22	91.99	99.10	91.10	83.65	90.63
Ours	90.03	92.62	99.28	91.31	84.00	90.93

1) Comparative Experiments on LEVIR-CD Dataset: We present the performance metrics of each comparative method on the LEVIR-CD dataset in Table I. In terms of OA, the performance of each method is relatively similar, with most achieving an OA value above 98%. This indicates that these methods are relatively mature, leaving limited room for further improvement. For instance, HRMNet achieves an OA of 98.99%, while our proposed method reaches 99.28%, demonstrating its superiority. Regarding the $F1$-score, DSAMNet achieves an $F1$-score of 84.43%, which is relatively low. This is primarily because it focuses solely on feature processing without addressing pseudochanges. In contrast, AANet, which employs an ARM to handle ambiguous regions in pseudochanges, achieves an $F1$-score of 90.94%. Our proposed method further improves this metric, reaching an $F1$-score of 91.31%. This improvement is attributed to our method's comprehensive handling of two types of pseudochanges, thereby reducing the interference of irrelevant changes on the model. IoU is used to measure the overlap degree between the predicted results and the actual results. Our method leads other methods with an IoU value of 84.00%. The IoU values of FC-Siam-diff, FC-Siam-conc, SRCD-Net, and DSAMNet are 76.16%, 76.75%, 75.21%, and 73.05%, respectively, all lower than 80%. Although these models involve deep feature extraction and some incorporate attention mechanisms, their robustness is still insufficient when dealing with complex scenarios.

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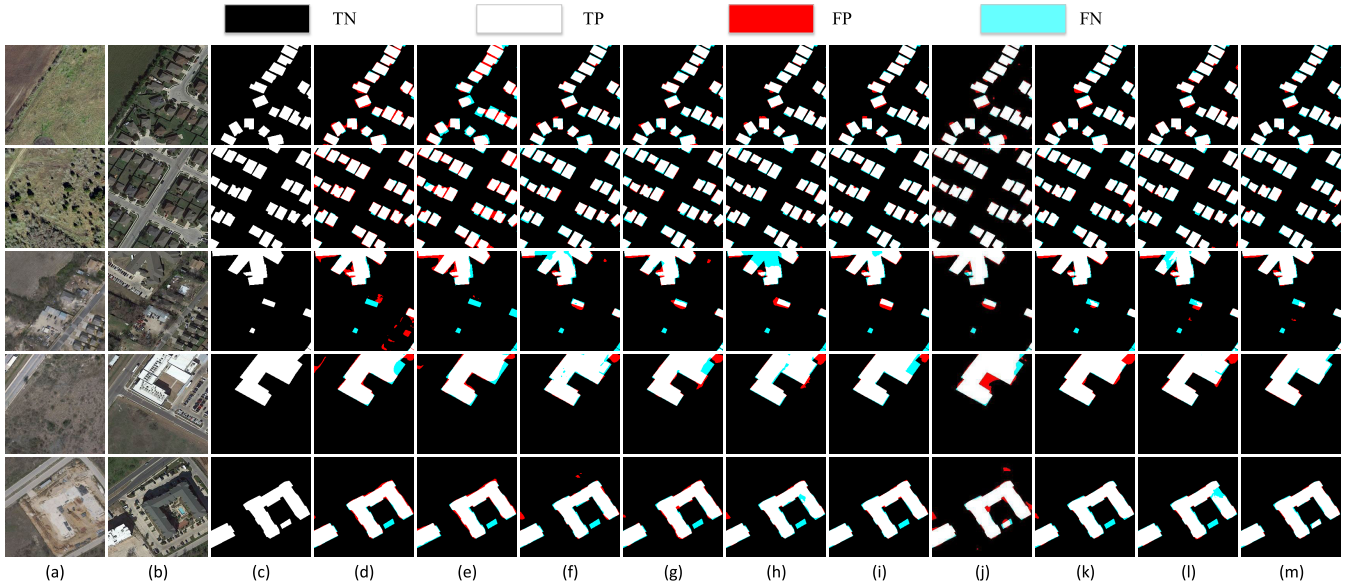


Fig. 5. Qualitative results of the overall methods on the LEVIR-CD dataset. The white, black, red, and cyan colors represent TP, TN, FP, and FN results, respectively. (a) Image T1. (b) Image T2. (c) GT. (d) SRCDNet. (e) DSAMNet. (f) BIT. (g) MSCANet. (h) DDCNN. (i) DMINet. (j) AANet. (k) HRMNet. (l) ScratchFormer. (m) Ours.

We present the quantitative results of multiple comparative methods on the LEVIR-CD dataset in Fig. 5. Each row represents a set of comparison images. In the first set, the T1-time image shows an open space, while the T2-time image exhibits dense buildings. Meanwhile, the GT clearly marks the building outlines. However, DSAMNet has poor recognition performance in the boundary areas and fails to identify the changed regions, with a large proportion of cyan areas. In contrast, our method effectively mitigates the interference of shadow areas, accurately identifies the changed parts, and highly coincides with the GT.

In the second set of comparison images, the area is an open space at time T1, and by time T2, it is covered with dense buildings that nearly occupy the entire image. In the results of SRCDNet and DSAMNet, there are relatively many red regions, indicating that nonchanged areas are misidentified as changed areas. Most of the other comparative methods predominantly show blue, suggesting an offset has occurred. Our method, having undergone feature registration, achieves more accurate boundary delineation.

In the third set of comparison images, at T1, only some buildings are present in the lower-right corner of the image. At T2, new buildings emerge in the upper part, and some scattered new buildings also appear in the lower-right corner. Some of the comparative methods fail to detect the newly appeared buildings in the upper part. This is because the area is relatively dark in color, making it liable to be misclassified as shadow. Our method largely overcomes this problem, and the CD results in this area almost match the GT. Virtually all of the comparative methods overlook the scattered buildings, but our method manages to detect them.

In the fourth set of comparison images, a building with a complex shape emerged in the image at time T2 compared to that at time T1. In the results of AANet, red regions appeared at the corners. Methods such as DSAMNet and SRCDNet misidentified the shadow area of the building as a changed area. In contrast, our method was not affected by the shadow area and achieved a relatively high accuracy in the detection results.

TABLE II

COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON SYSU-CD DATASET. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND GREEN

Methods	Rec(%)	Pre(%)	OA(%)	F1(%)	IoU(%)	KC(%)
FC-Siam-diff	58.18	90.53	88.70	70.84	54.84	64.24
FC-Siam-conc	72.92	80.44	89.43	76.49	61.94	69.70
SRCDNet	78.60	73.57	88.29	76.00	61.29	68.27
DSAMNet	82.16	74.88	89.29	78.35	64.41	71.26
BIT	76.19	81.90	90.41	78.94	65.21	72.74
MSCANet	75.70	81.33	90.17	78.42	64.49	72.06
DDCNN	78.88	81.82	90.89	80.32	67.12	74.40
DMINet	79.76	84.13	91.68	81.89	69.33	76.49
AANet	79.81	84.55	92.61	82.11	69.66	77.46
HRMNet	80.81	83.97	91.84	82.16	70.01	77.05
ScratchFormer	91.16	80.76	98.70	75.30	60.39	68.20
Ours	78.80	88.40	92.56	83.32	71.41	78.56

In the last set of comparison images, the image at time T1 shows the shape of the building's foundation, while the image at time T2 is a complete building. Due to the same shape of the building and the foundation, the detection difficulty is increased. Most of the comparative methods failed to identify the change of the smaller building in the middle. Moreover, the shadow on the edge part was not detected. However, with the suppression of the influence of shadow, our method detected the details and edges.

The comparisons of the above five groups of results all demonstrate the effectiveness and robustness of our method.

2) Comparative Experiments on SYSU-CD Dataset: We list the various metrics of multiple comparative methods on the SYSU-CD dataset in Table II. The performance of different comparative methods shows significant differences. However, the method we proposed still exhibits remarkable performance advantages on this dataset. Taking the $F1$ value as an example,

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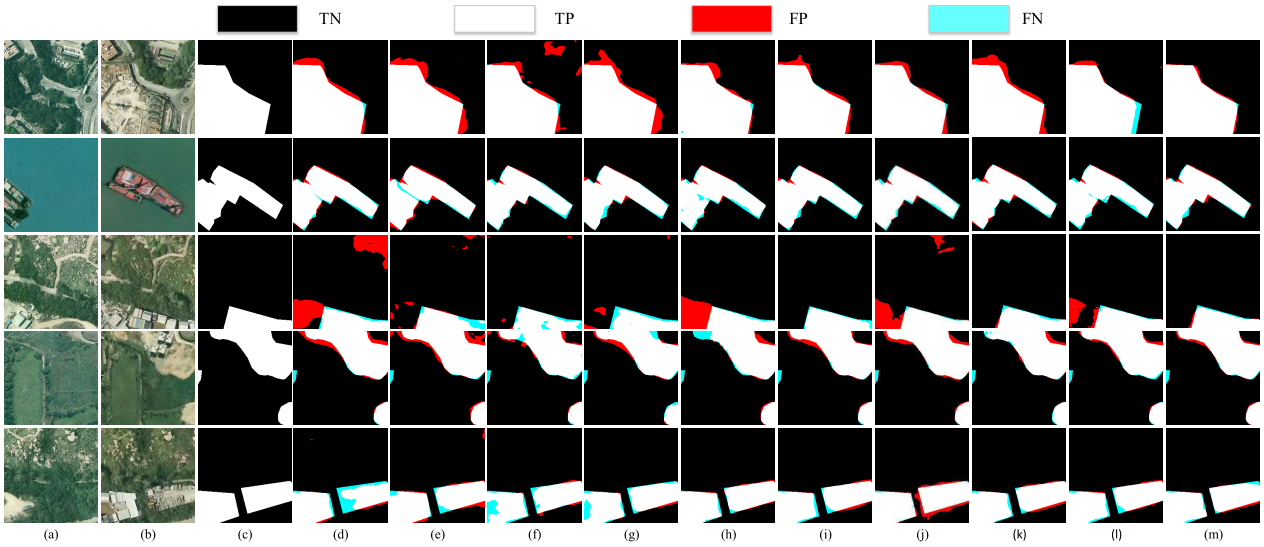


Fig. 6. Qualitative results of the overall methods on the SYSU-CD dataset. The white, black, red, and cyan colors represent TP, TN, FP, and FN results, respectively. (a) Image T1. (b) Image T2. (c) GT. (d) SRCDNet. (e) DSAMNet. (f) BIT. (g) MSCANet. (h) DDCNN. (i) DMINet. (j) AANet. (k) HRMNet. (l) ScratchFormer. (m) Ours.

among all the comparative methods, the HRMNet method has the highest $F1$ value, reaching 82.16%. Our method performs even better, with the $F1$ value soaring to 83.32%, an increase of 0.96% compared to the HRMNet method. In terms of IoU, the recently proposed DMINet method aligns the difference features at different levels through the Hadamard product operation, and its IoU value reaches 69.33%. Our method, with a more optimized algorithm and model design, boosts the IoU value to 71.41%, an increase of 2.08% compared to the DMINet method, significantly surpassing other comparative methods. Regarding the Kappa coefficient, which measures the consistency of classification, our method also performs outstandingly. Among the comparative methods, AANet, which has relatively prominent performance, has a Kappa value of 77.46%, while our method increases the Kappa value to 78.56%, an increase of 1.1%.

We present the quantitative results of multiple comparative methods on the SYSU-CD dataset in Fig. 6. Each row represents a set of comparison images.

In the first set of comparison images, a pentagonal building appears on the left side of the image at time T2 compared to time T1. The boundaries of the changed area are not distinct, posing difficulties for CD. As can be seen from Fig. 6, all comparative methods exhibit a certain degree of cyan in the changed area, indicating that this area has been undetected. In contrast, the undetected area of our method is minimal. Regarding the misdetection in the upper area, the performance of our method still needs improvement.

In the second set of comparison images, at time T1, there is a boat in the lower-left corner with water as the background. At time T2, a boat appears in the middle of the image, and the boat in the lower-left corner is gone. The background of this set of images is simple, making recognition relatively easy. Methods SRCDNet and DSAMNet show misdetection at the connection between the changes. The remaining methods only have a small amount of misdetection at the edges. Our method has the smallest undetected and misdetected areas.

In the third set of comparison images, the backgrounds of the images at both T1 and T2 are vegetated areas. At T2, a triangular building appears in the lower part of the image. SRCDNet and DDCNN misdetect the vegetation as a change. Our method mitigates the influence of the vegetation, thus avoiding misidentifying vegetation changes as real changes.

In the fourth set of comparison images, the changes from T1 to T2 are that the cultivated land in the upper part and the lower-right corner has turned into wasteland. The first two comparative methods have a relatively high number of undetected areas. This is because the edge regions are indistinct, and some are obstructed by shadows, increasing the difficulty of detection. The remaining comparative methods have more misdetections, which are present in the outlying areas. In contrast, our method has almost no undetected or misdetected areas.

In the last set of comparison images, the background of both the T1 and T2 images is vegetation. At T2, two building complexes appear in the lower part of the image. The dense vegetation surrounding the building complexes in the T2 image increases the difficulty of CD. It can be observed that most of the comparative methods have some misdetections or undetected areas. Only our method highly coincides with the GT. This is because the MSR-SS module we added can play a role in suppressing the influence of vegetation. Through the above quantitative comparisons, it is clear that our proposed method, RGCDNet, exhibits outstanding performance on the SYSU-CD dataset.

3) Comparative Experiments on WHU-CD Dataset: We list the various metrics of multiple comparative methods on the WHU-CD dataset in Table III. On this dataset, the method we proposed performs remarkably in all evaluation metrics.

In terms of recall, our method is close to the BIT method, but in other metrics, ours significantly outperforms BIT. For example, our IoU is 1.68% higher than that of BIT, indicating a higher overlap between our predicted change areas and the

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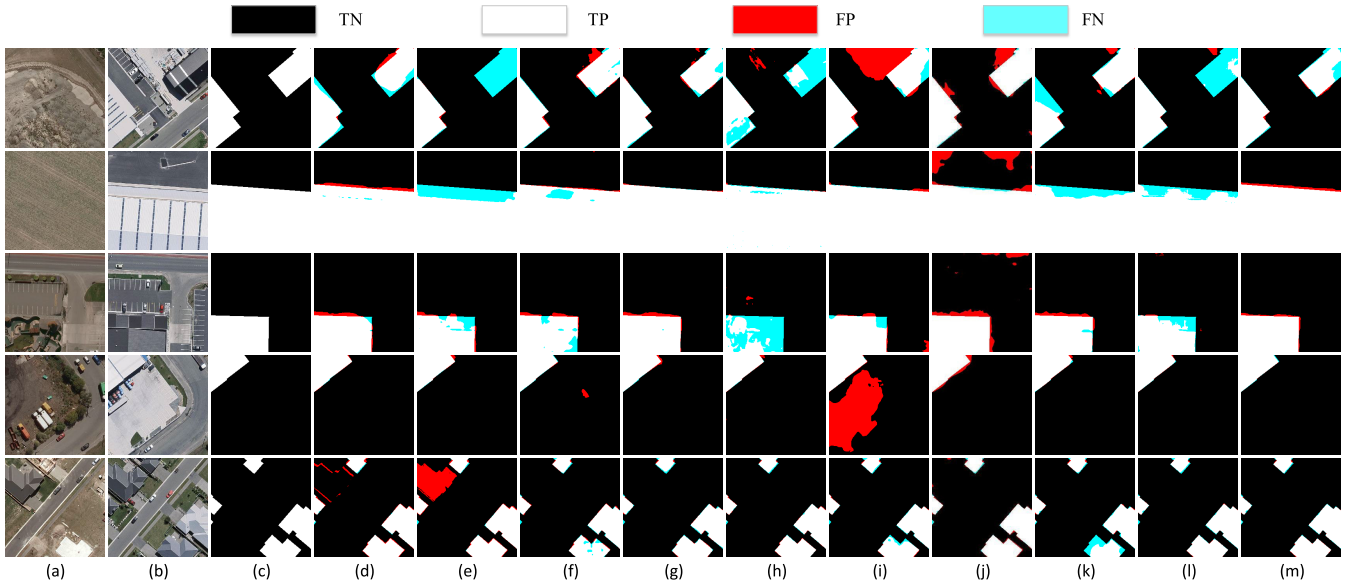


Fig. 7. Qualitative results of the overall methods on the WHU-CD dataset. The white, black, red, and cyan colors represent TP, TN, FP, and FN results, respectively. (a) Image T1. (b) Image T2. (c) GT. (d) SRCDNet. (e) DSAMNet. (f) BIT. (g) MSCANet. (h) DDCNN. (i) DMINet. (j) AANet. (k) HRMNet. (l) ScratchFormer. (m) Ours.

TABLE III
COMPARATIVE EXPERIMENT RESULTS FOR EACH COMPARISON METHOD ON WHU-CD DATASET. THE TOP TWO RESULTS ARE HIGHLIGHTED IN RED AND GREEN

Methods	Rec(%)	Pre(%)	OA(%)	F1(%)	IoU(%)	KC(%)
FC-Siam-diff	91.16	80.76	98.70	85.65	74.89	84.97
FC-Siam-conc	92.60	75.40	98.40	83.12	71.12	82.29
SRCDNet	91.90	89.51	99.19	90.69	82.96	90.27
DSAMNet	78.94	81.58	98.34	80.23	66.99	79.37
BIT	92.92	93.59	99.43	93.26	87.37	92.96
MSCANet	91.99	88.20	99.13	90.05	81.91	89.60
DDCNN	87.93	96.34	99.34	91.95	85.09	91.60
DMINet	89.19	87.02	98.97	88.09	78.72	87.55
AANet	89.84	79.69	98.62	83.64	73.18	82.93
HRMNet	87.15	92.70	99.16	89.84	81.55	89.40
ScratchFormer	89.91	94.76	99.36	92.28	85.66	91.94
Ours	93.11	95.45	99.54	94.21	89.05	93.87

actual ones. As most targets in the WHU-CD dataset are large-scale buildings, the CBAM of DSAMNet has limitations in handling edge information, resulting in lower IoU and $F1$ scores. The $F1$-score of our method reaches 94.21%. SRCDNet and DDCNN, with their strong feature extraction capabilities, achieve $F1$ scores of 90.69% and 91.95% respectively, which are 3.52% and 2.26% lower than our method, highlighting the comprehensive performance advantage of our method on this dataset.

We present the quantitative results of multiple comparative methods on the WHU-CD dataset in Fig. 7. Each row represents a set of comparison images.

In the first set of comparison images, the image at time T1 shows wasteland, and at time T2, two buildings appear, one in the lower-left corner and the other in the upper-right corner. DSAMNet fails to detect the change of the building in the upper-right corner. To varying degrees, other comparative methods misdetect the open space in the upper part as a changed area. Our method does not have such issues and highly coincides with the GT.

In the second set of comparison images, the image at T1 depicts a plain, while at T2, buildings emerge in the lower part of the image. The distinct difference between the edges and the interior of the buildings makes undetected areas likely to occur. For instance, methods like DSAMNet and BIT fail to detect the edges, and AAnet shows misdetections in the upper part of its result image. In contrast, our method does not have such issues and can accurately identify the changed areas.

In the third set of comparison images, the image at T1 appears dark, while the image at T2 is bright and has a square-shaped building added in the lower-left corner. Due to the color deviation, the open space above the changed area is prone to being misidentified as an actual change. All of the comparative methods exhibit both undetected and misdetected areas. Our method, however, can overcome the impact of illumination, accurately avoid the pseudochange areas, and significantly outperforms other methods.

In the fourth set of comparison images, there are no significant changes between the images at T1 and T2. The two black areas in the T2-time image are open spaces, which are easily misidentified as changed areas although no actual changes have occurred. Most of the comparative methods are misled by this area and exhibit misdetection. Our method, on the other hand, is not misled and demonstrates good adaptability.

In the last set of comparison images, compared to the image at T1, two new buildings appear at the upper-left and lower-right corners of the image at T2. SRCDNet misidentifies the shadow of the unchanged building in the upper-left corner as a changed area, and DSAMNet experiences a large-scale misdetection due to the impact of illumination changes. The results obtained by our method demonstrate strong anti-interference capabilities and almost coincide with the GT. Through the above comparisons, our method demonstrates a powerful anti-interference ability, as well as good adaptability to high-resolution images and multifeature changes.

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TABLE IV

ABLATION OF DIFFERENT MODULES ON THREE DATASETS. "VEGE" STANDS FOR VEGETATION SUPPRESSION, WHICH MITIGATES INTERFERENCE FROM VEGETATION, AND "WATER" REPRESENTS WATER BODY SUPPRESSION, DESIGNED TO DISTINGUISH WATER BODIES FROM SHADOWS

MSR-SS		CRM	LEVIR				SYSU				WHU			
Vege	Water		OA(%)	F1(%)	IoU(%)	KC(%)	OA(%)	F1(%)	IoU(%)	KC(%)	OA (%)	F1(%)	IoU(%)	KC(%)
			98.89	88.81	79.87	88.23	90.71	79.41	65.84	73.42	98.95	89.61	81.79	89.06
✓			99.26	90.92	83.36	90.54	91.05	79.48	65.95	73.76	99.04	90.09	81.96	89.58
	✓		98.92	89.38	80.80	88.81	90.81	79.95	66.60	74.00	99.26	90.63	82.87	90.25
✓	✓		99.26	91.10	83.66	90.71	91.99	82.60	70.40	77.40	99.52	93.97	88.63	93.73
		✓	99.17	89.82	81.52	89.39	90.84	80.28	67.05	74.31	99.26	91.13	83.71	90.75
✓	✓	✓	99.28	91.31	84.00	90.93	92.56	83.32	71.41	78.56	99.54	94.10	88.87	93.87

Considering the test results of the three datasets, the method we proposed can identify various changes in different scenarios. Whether it is affected by pseudochange interferences such as illumination, shadows, or geometric registration errors, our method demonstrates outstanding performance, strong adaptability, and excellent anti-interference capabilities.

In set (a), shadows cast by buildings are prone to being misclassified as boundaries of changed objects. After the MSR-SS module suppresses the influence of shadows, these regions appear as light blue in the feature maps, indicating minimal weights. Correspondingly, in the result maps, shadow regions are not identified as changed areas. Similar effects are observed in the remaining three sets, where shadow regions are assigned low weights, ensuring that the final detection results are not influenced by shadows, thereby enhancing the accuracy of CD.

E. Ablation Experiments

In order to thoroughly investigate the specific contributions of each module to the model performance, we designed and conducted ablation experiments on three datasets, using four evaluation metrics, namely OA, $F1$, IoU, and KC. The results are presented in Table IV, where a checkmark indicates that the module has been incorporated into the model. We analyzed the performance of the model under different configurations by gradually removing or retaining the proposed modules, thereby clarifying the importance and effectiveness of each module.

1) Effects of MRS-SS:

a) Effects of MRS-SS with only vegetation suppression: The first group of experiments retains only the vegetation interference suppression function, removing the water body interference processing part of the MSR-SS module as well as the entire CRM. This setting aims to examine the impact of the vegetation interference suppression function on shadow suppression, and consequently on model performance, thereby verifying the model's ability to extract and analyze target information when dealing with vegetation interference shadows.

As shown in Table IV, even though the MSR-SS module is not complete and only the vegetation suppression function is retained, our model still outperforms the baseline model on the three datasets. This indicates that addressing the confusion between vegetation and shadows can enhance the separability of shadows, thereby improving the accuracy of CD.

b) Effects of MRS-SS with only water suppression: The second group of experiments retains only the water body interference suppression function, removing the vegetation interference suppression part of the MSR-SS module as well as the CRM. This set of experiments focuses on the role of the water body interference suppression function in the model and explores the model's performance in relevant tasks when only dealing with the shadow of water body interference.

As shown in Table IV, when the model only deals with water body interference, its performance on the LEVIR dataset is slightly lower than that of the model only handling vegetation interference; however, it is slightly higher on the WHU dataset. This is attributed to the scarcity of samples containing water bodies in the dataset, which means that solely addressing water body interference does not significantly enhance the model. Nevertheless, it still outperforms the baseline model overall.

c) Effects of the complete MSR-SS: The third group of experiments retains the entire MSR-SS module while removing the CRM. This experimental group aims to evaluate the contribution of the complete MSR-SS module.

When all components of the MSR-SS module are retained in the model, there is a significant improvement in performance. For instance, on the LEVIR dataset, compared to the baseline model, the $F1$-score is increased by 2.29%, and the IoU is enhanced by 3.79%. This indicates that each process within the MSR-SS module is essential. The suppression of interference from vegetation and water bodies can greatly increase the separability of shadows, thereby improving the accuracy and robustness of detection.

Fig. 8 illustrates the impact of the MSR-SS module on shadow regions. The first row of Fig. 8 presents the images containing shadows in either I_A or I_B. The second row displays the visualization of the intermediate feature maps, which has been processed by the MSR-SS module. The third row shows the visualization of the final result maps. Four sets of images are presented in total, with red boxes highlighting the locations where shadows appear.

2) Effects of CRM: The fourth group of experiments retains only the CRM while removing the entire MSR-SS module. This setting is used to investigate the impact of the CRM alone on the model's performance.

As shown in Table IV, compared to the third group with only the MSR-SS module, the overall performance of this setting has slightly decreased, while it is slightly better than the second group. Compared to the baseline model, there are significant improvements in all metrics. For example, on the WHU-CD dataset, the $F1$-score increased by 1.52%, and the IoU increased by 1.92%. The above results indicate that the CRM plays a key role in the registration task.

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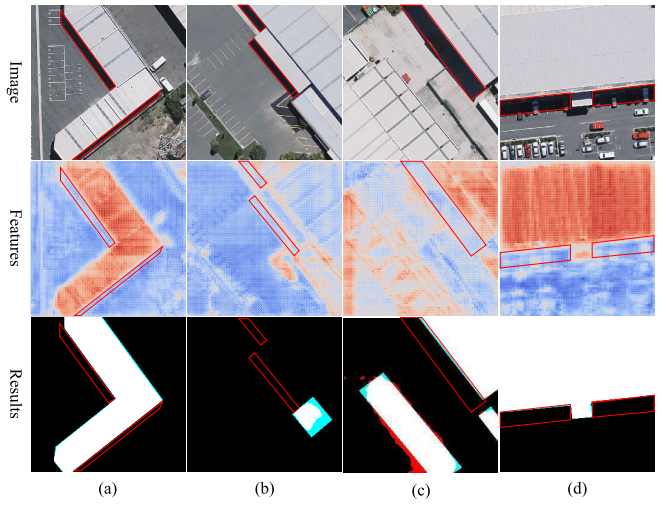


Fig. 8. Visualization results of feature maps after processing by MSR-SS on the WHU-CD dataset. (a) Scenario 1: detailed visualization of shadow suppression for a large-scale building with regular geometry. (b) Scenario 2: shadow detection and suppression results for complex building structures and associated shadows. (c) Scenario 3: results showing how the MSR-SS module handles shadows that could be easily misclassified as changed boundaries. (d) Scenario 4: visualization of the module's effectiveness in distinguishing between real changes and pseudo-changes caused by shadows.

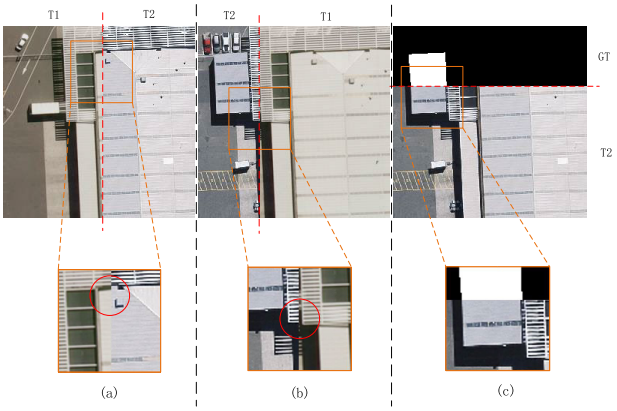


Fig. 9. Visual comparison of the registration refinement effects achieved by the proposed CRM module. (a) Stitching result between the T1 and T2 images. (b) Stitching result between the T2 and T1 images. (c) Stitching result between the T2 image and our CRM-refined output.

We designed the comparative visualization shown in Fig. 9. Fig. 9(a) and (b) first establishes the existence of inherent registration errors in the original image pair from two perspectives. Fig. 9(c) then clearly demonstrates the output of our CRM module as the solution—the previous misalignments are largely eliminated. This clear 'problem-solution' visualization starkly illustrates the core contribution of the CRM module.

3) Full Model: The fifth group of experiments retains all the complete modules. This experimental group aims to evaluate the contribution of the synergistic effect of the two modules proposed in our study.

As shown in Table IV, the complete model outperforms all other experimental groups in terms of all evaluation metrics, which fully demonstrates the significant improvement of the MSR-SS and CRM modules on CD performance. The synergistic effect of these two modules is crucial for enhancing the precision of CD, highlighting their indispensability in the model.

TABLE V

PERFORMANCE COMPARISON ($F1$ AND IOU IN %) WITH DIFFERENT VALUES OF HYPERPARAMETER α ON THE LEVIR-CD, WHU-CD, AND SYSU-CD DATASETS. THE BEST RESULTS FOR EACH DATASET ARE HIGHLIGHTED IN BOLD

α	LEVIR-CD		WHU-CD		SYSU-CD	
	F1	IoU	F1	IoU	F1	IoU
0.3	90.02	83.34	92.82	86.60	82.02	68.84
0.4	91.11	83.28	93.95	88.59	82.91	70.53
0.5	**91.31**	**84.00**	**94.21**	**89.05**	**83.32**	**71.41**
0.6	89.86	82.76	94.04	88.76	81.86	70.77

F. Discussion on the α of Loss Function

In our previous research, we introduced a composite loss function, as shown in (21), where α serves as a key hyperparameter. To thoroughly analyze the impact of α on the model's generalization ability, we extended our initial analysis on the WHU-CD dataset to include two additional public benchmarks, LEVIR-CD and SYSU-CD. We evaluated the performance with α set to 0.3, 0.4, 0.5, and 0.6.

The comprehensive results are summarized in Table V. Critically, the optimal value of $\alpha = 0.5$ is consistent across all three datasets, achieving the highest $F1$ and IoU scores. On the WHU-CD dataset, $\alpha = 0.5$ yielded an $F1$-score of 94.21% and an IoU of 89.05%, outperforming other values. This trend holds on the diverse scenarios represented by LEVIR-CD ($F1$: 91.31%, IoU: 84.00%) and SYSU-CD ($F1$: 83.32%, IoU: 71.41%).

The consistency in these results demonstrates that $\alpha = 0.5$ provides the optimal balance for our loss function, robustly generalizing across different geographical distributions and change characteristics (e.g., from urban to rural settings). This strongly suggests that the parameter choice is not overfitted to a specific data distribution but is a reliable default for the proposed method.

While the recall rate on WHU-CD alone suggested a peak at $\alpha = 0.5$ (93.11%, compared to 92.38% at $\alpha = 0.4$), the multidataset analysis of the more holistic $F1$ and IoU metrics provides a much stronger foundation for this conclusion. The decline in performance when α deviates from 0.5 (e.g., to 0.6) across all datasets confirms that an imbalance in the loss function's composition can impair the model's detection and localization capabilities.

Overall, based on extensive validation across three standard datasets, we conclude that $\alpha = 0.5$ is the robust and optimal setting for our proposed loss function, ensuring its effectiveness and generalizability.

G. Comparison of Parameters and Time

Beyond the accuracy metrics, the computational efficiency and practicality of a model are crucial for its real-world deployment. In this section, we present a comprehensive comparison of the computational cost and inference speed between the proposed method and other state-of-the-art approaches.

Table VI summarizes the parameter count, computational complexity (in GFLOPs), average inference time, and the mIoU for all competing methods. Our method achieved the highest $F1$ and IoU scores among all listed methods. In terms

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TABLE VI
COMPARISON OF PARAMETER COUNT, COMPUTATIONAL COMPLEXITY, AND INFERENCE TIME ON THE FULL LEVIR-CD TEST SET

Methods	Params (M)	FLOPs (G)	Time (ms)	F1 (%)	IoU (%)
FC-Siam-diff	1.35	4.72	12	86.47	76.16
FC-Siam-conc	1.55	5.32	12	86.84	76.75
SRCDNet	16.98	71.68	22	85.85	75.21
DSAMNet	16.95	75.39	32	84.43	73.05
BIT	11.47	26.31	13	89.50	80.99
MSCANet	16.42	14.80	16	89.75	81.40
DDCNN	46.68	177.44	76	90.60	82.82
DMINet	6.24	14.55	29	90.78	83.12
AANet	15.82	24.21	23	90.77	83.13
HRMNet	13.44	11.54	14	89.92	81.69
ScratchFormer	36.95	196.59	73	91.10	83.65
Ours	**17.43**	**44.74**	**18**	**91.31**	**84.00**

of efficiency, the total inference time for the proposed method on the complete test set was 18 s. This duration is longer than that of the fastest models (FC-Siam-diff and FC-Siam-conc at 12 s) but is shorter than that of other high-accuracy models such as ScratchFormer (73 s) and DDCNN (76 s). The parameter count of the proposed model (17.43 M) is moderate compared to others.

The results indicate that the performance gain of our model is not merely a result of increased model capacity. The architectural design, particularly the CRM, is computationally efficient. In conclusion, while our parameter count is not the absolute lowest, the proposed method establishes a new state-of-the-art in performance while maintaining a very efficient inference speed.

V. CONCLUSION

In this article, we present a novel CD method, RGCD-Net, designed to address the two major pseudochange issues in remote sensing images: shadows and geometric registration errors. These improvements enhance detection accuracy and model robustness. Our approach integrates multisource radiometric feature analysis with deep learning techniques to develop the MSR-SS and CRM modules. The MSR-SS module processes bitemporal remote sensing images by extracting and channel-splitting image features. It leverages the high reflectance of vegetation in the NIR band and the low reflectance of shadows, as well as the distinct reflectance characteristics of inland water bodies in the blue and green bands compared to the high blue-band reflectance of shadows. Through vegetation and water body interference suppression steps, the module enhances the separability of shadows from other land features. Subsequently, shadow feature transformation and a shadow attention mechanism are applied to identify and suppress shadows effectively. After the encoder extracts features at different scales, these features are passed to the CRM module. Within CRM, registration is integrated into the decoder. The module predicts image deformation fields and combines them with the CD decoder. Using bidirectional deformation and weighted fusion of deformed and undeformed images, CRM achieves collaborative optimization of registration and CD, effectively reducing registration errors. The results of numerous ablation experiments have proven the significant contributions of each key component in the model to the overall performance. Comparative experiments on three public datasets have demonstrated that our model outperforms traditional methods and other advanced models in various evaluation metrics.

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